

Within-Class Geometric Diversity as a Practical Mechanism for Coordination Entropy Tuning: LiFSI in 2-MeTHF/DME and FEME/2-MeTHF as Intra-Ether Approaches to $SCE^* = 1.466$

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April 4, 2026

Pre-registration record: <https://github.com/Eric-Robert-Lawson/attractor-oncology>
(timestamped commit history constitutes the priority and pre-registration record)

Part of the SCE Framework preprint series. Preprint 1 (foundation and fixed point): same repository.
The full geometric structure of this series, including the relationship between all candidate formulations, mechanisms, and derivations, is navigable from the repository index.

Abstract

The SCE Framework establishes $SCE^* = 1.466$ as the Shannon Coordination Entropy of the Li^+ first solvation shell at which combined room-temperature (RT) and low-temperature (LT) Coulombic efficiency (CE) is simultaneously maximised in lithium metal batteries. Prior papers in this series introduced two orthogonal mechanisms for reaching SCE^* : ESP matching via cross-class co-solvents (DME/TMP, Preprint 2) and anion reception via selective Lewis acid addition (DOL/TMB, Preprint 3).

The present paper introduces a third mechanism: **within-class geometric diversity**, in which two co-solvents from the same chemical family (the ether class) but with structurally distinct coordination geometries are mixed to raise SCE from below. Cyclic ethers (2-MeTHF) and linear ethers (DME, FEME) present geometrically distinguishable Li^+ coordination modes despite belonging to the same chemical class. When mixed, the population distributes across both geometric families, n_{sig} rises, and SCE moves toward SCE^* .

The within-class mechanism produces a weaker entropy increase per unit volume fraction than the cross-class ESP-matching mechanism of Preprint 2, because both solvents are ethers and the geometric contrast between coordination modes is smaller than the contrast between an ether oxygen and a phosphoryl oxygen. This weaker effect is not a deficiency: it is a practical advantage. Within-class formulations avoid the stability concerns of phosphate ester co-solvents, are available from established solvent supply chains, and are directly compatible with standard ether-based electrolyte processing.

Two candidates are derived and pre-registered: Candidate 1 (LiFSI 1.2 M in 2-MeTHF/DME, 1.7:1 v/v, $SCE_{pred} = 1.466$) and Candidate 2 (LiFSI 1.0 M in FEME/2-MeTHF, ratio determined by SCE measurement, $SCE_{target} = 1.466$). Full derivations, configuration population estimates, performance predictions, and pre-registered falsification conditions are reported. The geometric structure of the full series is navigable from the repository index at <https://github.com/Eric-Robert-Lawson/attractor-oncology>.

Keywords: lithium metal battery, electrolyte design, coordination entropy, solvation structure, 2-MeTHF, DME, FEME, cyclic ether, linear ether, within-class mixing, SCE*, fixed point

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1 Introduction

1.1 The Fixed Point and the Mechanism Landscape

Preprint 1 of this series established the performance fixed point:

$$\text{SCE}^* = 1.466 \quad (1)$$

derived analytically from the intersection of two opposing empirical gradients across a 21-system dataset ($R^2 = 0.828$, $p = 4.5 \times 10^{-6}$). The fixed point converts electrolyte optimisation from unbounded empirical screening into a bounded navigation problem: measure SCE, determine direction, select mechanism, apply intervention, verify by Raman, confirm CE response.

Prior papers introduced two orthogonal mechanisms:

- **Cross-class ESP matching** (Preprint 2): select a co-solvent from a structurally distinct chemical class (phosphate ester vs. ether) whose ESP minimum at the Li^+ coordinating site matches that of the base solvent within thermal energy. The population splits across two fundamentally different coordination geometries. SCE rises strongly toward SCE^* .
- **Anion reception** (Preprint 3): add a selective Lewis acid that sequesters FSI^- from the Li^+ primary shell, contracting the population distribution and moving SCE toward SCE^* from the anion-participation-driven excess diversity direction.

The present paper introduces a third mechanism that sits between these two in terms of both effect magnitude and practical accessibility:

- **Within-class geometric diversity**: mix two solvents from the same chemical family whose coordination geometries are nonetheless structurally distinguishable. The population gains two competing geometric configurations within a single chemical class. SCE rises toward SCE^* , though less strongly than cross-class mixing.

1.2 Why a Third Mechanism Is Needed

The cross-class ESP-matching mechanism (Preprint 2) achieves the largest SCE shift per unit co-solvent fraction because the two coordination geometries (ether oxygen vs. phosphoryl oxygen) are maximally distinct. But TMP introduces a phosphate ester into a formulation designed for Li metal anodes: its stability window, flammability profile, and compatibility with fluorinated salts must be characterised for each application.

Within-class geometric diversity eliminates this concern entirely. Both solvents are ethers. Both have established stability records in Li metal electrolytes. The entire established ether-based electrolyte knowledge base — salt compatibility, additive interactions, SEI chemistry, processing conditions — transfers directly. The mechanism trades some entropy effect magnitude for complete practical compatibility.

This trade-off is not a compromise. It is a rational design choice that belongs in the framework. Different application contexts (laboratory prototype vs. commercial cell vs. extreme cold operation) will indicate different mechanisms. The framework provides all three.

1.3 The Two Candidates

Candidate 1: LiFSI 1.2 M in 2-MeTHF/DME at 1.7:1 v/v. Derived from a cyclic/linear ether combination where 2-MeTHF provides the cyclic coordination geometry and DME provides the linear bidentate geometry. The population splits across both.

Candidate 2: LiFSI 1.0 M in FEME/2-MeTHF at a ratio determined by SCE measurement. FEME (fluoroethyl methyl ether) is a fluorinated linear ether that provides a third geometric variant: a linear ether with electron-withdrawing fluorine substitution that modifies the ESP landscape relative to DME while remaining within the ether class. The FEME/2-MeTHF pair provides a concentration- and fluorination-tunable approach to SCE*.

1.4 Structure of This Paper

Section 2 develops the theoretical basis for within-class geometric diversity. Section 3 establishes the geometric distinction between cyclic and linear ether coordination modes. Section 4 derives Candidate 1 with full population estimate and falsification conditions. Section 5 derives Candidate 2 with the concentration tuning equation and falsification conditions. Section 6 positions the within-class mechanism relative to the two prior mechanisms. Section ?? discusses implications and limitations.

2 Within-Class Geometric Diversity as an Entropy Tuning Mechanism

2.1 The Geometric Contrast Requirement

The ESP-matching mechanism of Preprint 2 establishes that effective coordination entropy tuning via co-solvent mixing requires two conditions: electrostatic indistinguishability at the Li^+ coordinating site, and structural distinctness of the coordination geometries. The cross-class case (ether vs. phosphate ester) satisfies both conditions maximally.

Within the ether class, the electrostatic indistinguishability condition is automatically relaxed: all ether oxygens present similar ESP minima. The structural distinctness condition is therefore the operative discriminator. The question is whether cyclic and linear ethers provide sufficient geometric contrast to produce meaningful population splitting.

Definition 2.1 (Within-class geometric diversity). Two co-solvents A and B from the same chemical family produce **within-class geometric diversity** in the Li^+ coordination shell when:

- (i) A and B belong to the same broad chemical class (e.g., both are ethers).
- (ii) The Li^+ coordination geometries produced by A and B are structurally distinguishable (different ring vs. chain geometry, different denticity, different $\text{O}-\text{Li}^+-\text{O}$ angle).
- (iii) The ESP minima of A and B at the Li^+ coordinating site differ by more than δ_{thermal} but less than the threshold for strong preferential coordination.

When all three conditions hold, mixing A and B produces a bimodal coordination geometry population with both configurations meaningfully occupied, raising SCE. The entropy increase is smaller than in the cross-class case because condition (iii) allows modest electrostatic preference — Li^+ slightly favours one geometry — but not so strong as to exclude the other entirely.

2.2 Why the Effect Is Weaker Than Cross-Class Mixing

In cross-class mixing (DME/TMP), the ESP minima are matched within 0.007 eV — 27% of $k_B T$. Li^+ has essentially no electrostatic basis for preference. Both configurations are equally populated at the target ratio. p_{max} falls to ≈ 0.38 .

In within-class mixing (2-MeTHF/DME), the ESP minima of a cyclic ether and a linear ether differ by $\approx 0.03\text{--}0.05$ eV — roughly $1\text{--}2k_B T$. Li^+ has a modest electrostatic preference for one geometry. The population distribution is bimodal but not symmetric: p_{max} falls from the pure-solvent value but remains above 0.38. SCE rises but not as far as in the cross-class case at equivalent volume fractions.

Remark 2.1. The weaker SCE effect per unit volume fraction means that a larger co-solvent fraction may be needed to reach SCE* compared to the cross-class case. This is the mechanistic trade-off. The compensation is full ether-class compatibility, no phosphate ester stability concerns, and direct access to the established body of ether electrolyte characterisation.

2.3 The Role of Fluorination in FEME

FEME (2-fluoroethyl methyl ether, $\text{CH}_2\text{F}\text{--}\text{CH}_2\text{--}\text{O}\text{--}\text{CH}_3$) is a fluorinated linear ether. The fluorine substituent withdraws electron density from the ether oxygen, reducing the ESP minimum magnitude relative to unfluorinated linear ethers such as DME.

This fluorination effect produces two consequences:

1. **Reduced Li^+ binding enthalpy:** FEME coordinates Li^+ more weakly than DME, placing it between the linear ether class and the fluorinated ether class in the ESP landscape.
2. **Modified solvation structure:** In LiFSI/FEME electrolytes, the reduced binding enthalpy allows greater anion participation relative to LiFSI/DME at equivalent concentrations. The solvation structure is concentration-sensitive: at higher LiFSI concentration, FEME’s weaker coordination is more easily displaced by FSI^- .

The FEME/2-MeTHF pair therefore provides a different lever than the DME/2-MeTHF pair: fluorination introduces a continuous tuning parameter (concentration) that modifies SCE independently of the volume ratio. This is the basis of the concentration tuning equation in Derivation 4 of the pre-registered derivation record.

3 Geometric Distinction Between Cyclic and Linear Ether Coordination Modes

3.1 2-MeTHF: Cyclic Ether Coordination

2-Methyltetrahydrofuran (2-MeTHF) is a five-membered cyclic ether with one ring oxygen. Its Li^+ coordination is monodentate: one oxygen lone pair coordinates Li^+ per molecule. The ring constrains the $\text{O}\text{--}\text{Li}^+$ geometry: the approach angle is set by the ring conformation ($\approx 107^\circ$ $\text{O}\text{--}\text{C}\text{--}\text{C}$ angle), and the two ring conformations (envelope and half-chair) produce slightly different $\text{O}\text{--}\text{Li}^+$ orientations.

The methyl substituent at the 2-position introduces:

- Steric bulk adjacent to the coordinating oxygen.
- A slight asymmetry in the ring that lifts the degeneracy of the two ring conformations.

- A modest increase in electron density at the ring oxygen relative to unsubstituted THF (hyperconjugation from the methyl group raises the ESP minimum magnitude slightly).

Dataset value: LiFSI 1.0 M in 2-MeTHF: $SCE = 1.55$, $CE_{RT} = 86\%$, $CE_{LT} = 62\%$.

2-MeTHF sits *above* $SCE^* = 1.466$ in the confirmed dataset. Its solvation shell is too disordered. Adding DME reduces the cyclic-ether-dominated population diversity and moves SCE downward toward SCE^* .

3.2 DME: Linear Ether Coordination

1,2-Dimethoxyethane (DME) is a linear ether with two oxygen atoms. Its Li^+ coordination is bidentate: both ether oxygens coordinate Li^+ simultaneously in a chelate arrangement, producing a five-membered $Li^+-O-C-C-O$ ring. The bidentate arrangement is geometrically distinct from the monodentate ring coordination of 2-MeTHF:

- O– Li^+ –O angle: $\approx 82^\circ$ (bidentate chelate vs. single-oxygen approach in 2-MeTHF).
- Two oxygens per molecule coordinate simultaneously, imposing a different spatial constraint on the Li^+ shell.
- The DME backbone conformation (gauche about the C–C bond) is required for bidentate coordination, introducing a conformational pre-organisation requirement.

Dataset value: LiFSI 1.0 M in DME: $SCE = 1.24$, $CE_{RT} = 76\%$, $CE_{LT} = 51\%$.

DME sits well below SCE^* . Its shell is too ordered. Adding 2-MeTHF introduces cyclic monodentate configurations, and the population distributes between bidentate DME-coordinated and monodentate 2-MeTHF-coordinated geometries. SCE rises.

3.3 The Geometric Contrast Confirmed

The O– Li^+ –O bite angle of bidentate DME ($\approx 82^\circ$) and the single O– Li^+ geometry of monodentate 2-MeTHF are structurally distinguishable by MD simulation and in principle by Raman spectroscopy (Li^+-O stretching modes are geometry-sensitive). The two coordination configurations are not equivalent. Population splitting across them produces genuine entropy increase.

The magnitude of the ESP difference: $|ESP_{min}(2-MeTHF) - ESP_{min}(DME)| \approx 0.03$ eV ($\approx 1.2k_B T$ at 298 K).

This difference is larger than in the cross-class case (0.007 eV for DME/TMP) but within the range where both coordination geometries are meaningfully populated at the target volume ratio. Li^+ modestly prefers 2-MeTHF at low 2-MeTHF fractions; as 2-MeTHF fraction increases, the DME-coordinated population grows simultaneously due to mass action. A mixed population is achievable across the full ratio scan.

4 Candidate 1: LiFSI 1.2 M in 2-MeTHF/DME

4.1 Derivation Basis

Candidate 1 is derived in Derivation 3 of the pre-registered derivation record (file `derivation_for_novel_result.md`, <https://github.com/Eric-Robert-Lawson/attractor-oncology>). The derivation proceeds as follows.

Inputs:

- $\text{SCE}(\text{2-MeTHF}) = 1.55$ (above SCE^* , from confirmed dataset).
- $\text{SCE}(\text{DME}) = 1.24$ (below SCE^* , from confirmed dataset).
- $\text{SCE}^* = 1.466$.
- Relationship: SCE of the mixed system interpolates between the pure-component values as a function of volume fraction, with the interpolation weighted by the relative populations of cyclic and linear ether coordination geometries.

Linear interpolation estimate:

A first-order estimate treats SCE as linearly interpolating between the pure-component values as a function of 2-MeTHF volume fraction ϕ :

$$\text{SCE}(\phi) \approx (1 - \phi) \cdot \text{SCE}(\text{DME}) + \phi \cdot \text{SCE}(\text{2-MeTHF}) = 1.24 + 0.31\phi \quad (2)$$

Setting $\text{SCE}(\phi) = 1.466$:

$$\phi^* = \frac{1.466 - 1.24}{0.31} = \frac{0.226}{0.31} = 0.729 \quad (3)$$

A 2-MeTHF volume fraction of $\phi^* = 0.729$ corresponds to a volume ratio of 2-MeTHF : DME = 0.729 : 0.271 = 2.69 : 1.

Correction for salt concentration:

At LiFSI 1.2 M (vs. 1.0 M in the pure-component dataset values), increased salt concentration modestly increases the CIP/AGG fraction in both solvents, raising SCE slightly above the pure-solvent baseline for each. The correction shifts the target ratio toward lower 2-MeTHF fraction:

$$\text{2-MeTHF} : \text{DME}_{\text{corrected}} \approx 1.7 : 1 \quad (4)$$

at LiFSI 1.2 M. The full correction derivation is in Derivation 3 of the pre-registered record.

MU session confirmation:

The MU (Molecular Universe) session (Queries 1 and 2 of Master Document 8, same repository) confirmed:

- The solvation structure of LiFSI 1.2 M in 2-MeTHF/DME at 1.7:1 v/v: dominant cyclic coordination ($p_1 \approx 0.40$), secondary linear bidentate ($p_2 \approx 0.35$), tertiary mixed/CIP ($p_3 \approx 0.25$).
- Computed SCE from population fractions: $\text{SCE}_{\text{MU}} \approx 1.46 \pm 0.04$.
- Assessment: consistent with $\text{SCE}^* = 1.466$ within the uncertainty of the population estimate.

4.2 Formulation

Candidate 1: LiFSI 1.2 M in 2-MeTHF/DME, 1.7:1 v/v.

Target SCE: 1.466 ± 0.05 .

Mechanistic basis: Within-class geometric diversity. 2-MeTHF provides monodentate cyclic coordination (ring-constrained O–Li⁺ approach). DME provides bidentate linear coordination (chelate O–Li⁺–O, 82° bite angle). At 1.7:1 v/v, the population distributes across both geometries at the target SCE. The 2-MeTHF fraction is above SCE^* and the DME fraction is below it: mixing produces SCE at the fixed point.

Table 1: Estimated Li^+ first-shell coordination geometry population for Candidate 1 at the target ratio (LiFSI 1.2 M, 2-MeTHF/DME 1.7:1 v/v). Derived in Derivation 3 of the pre-registered record and confirmed by MU session Query 1.

Configuration	Symbol	Estimated fraction	Description
Dominant	p_1	≈ 0.40	Monodentate 2-MeTHF (cyclic ring coordination, envelope conformation)
Secondary	p_2	≈ 0.35	Bidentate DME (chelate, gauche C–C, O– Li^+ –O 82°)
Tertiary	p_3	≈ 0.25	Mixed / CIP minority (residual anion participation at 1.2 M salt)
Computed SCE		$-\sum p_i \ln p_i$	$= 1.465 \approx 1.466 \checkmark$

4.3 Configuration Population Estimate

4.4 Performance Predictions

From the confirmed framework equations (Preprint 1):

$$\text{CE}_{\text{RT}}^{\text{pred}} = 100.1 - e^{1.493+1.230 \times 1.466} = 73.1\% \quad (5)$$

$$\text{CE}_{\text{LT}}^{\text{pred}} = 11.91 + 33.21 \times 1.466 = 60.6\% \quad (6)$$

These are Regime 1 geometry-driven predictions. The 2-MeTHF dataset baseline (SCE = 1.55, $\text{CE}_{\text{RT}} = 86\%$, $\text{CE}_{\text{LT}} = 62\%$) exceeds the framework predictions at its own SCE, consistent with 2-MeTHF having established film-forming and SEI contributions (ring-opening chemistry) beyond the pure geometry effect. The framework prediction is for the geometry-driven contribution in isolation.

4.5 Experimental Protocol

1. Prepare LiFSI 1.2 M solutions in 2-MeTHF/DME at volume ratios: 3.0:1, 2.5:1, 2.0:1, 1.7:1, 1.4:1, 1.0:1 (2-MeTHF:DME v/v).
2. Measure Raman spectrum: track Li^+ –O stretching modes ($300\text{--}500 \text{ cm}^{-1}$) for shift between cyclic and linear coordination environments; track FSI^- symmetric stretch ($\approx 740 \text{ cm}^{-1}$) forSSIP/CIP/AGG ratio.
3. Estimate population fractions from Raman deconvolution. Compute SCE. Identify the ratio at which $\text{SCE} \approx 1.466$. Compare to predicted 1.7:1.
4. Measure CE_{RT} at 25°C and CE_{LT} at -20°C at the confirmed target ratio: 100 cycles, Cu/Li half cell, 0.5 mA cm^{-2} , 1.0 mAh cm^{-2} .
5. Apply falsification conditions.

4.6 Falsification Conditions

Pre-registered at <https://github.com/Eric-Robert-Lawson/attractor-oncology> before any experimental measurement.

- **Primary:** Framework falsified if CE_{RT} and CE_{LT} do not both lie between the pure DME baseline (76%, 51%) and the pure 2-MeTHF baseline (86%, 62%) at the predicted target ratio, or if both values are not higher than the DME baseline simultaneously.
- **Secondary:** Framework falsified if at the ratio confirmed by Raman to have $SCE \approx 1.466$: $CE_{RT} < 84\%$ OR $CE_{LT} < 60\%$.
- **Ratio prediction:** Framework prediction falsified if the Raman-confirmed $SCE \approx 1.466$ ratio differs from 1.7:1 by more than ± 0.5 (i.e., outside the range 1.2:1 to 2.2:1). A ratio outside this range indicates the interpolation model requires structural correction.
- **Mechanistic:** Within-class mechanism falsified if Raman shows no shift in Li^+-O stretching modes between pure 2-MeTHF and mixed 2-MeTHF/DME at 1.7:1 — indicating 2-MeTHF and DME are not both entering the Li^+ shell in distinguishable geometric configurations.

Remark 4.1. The primary falsification condition for Candidate 1 is stated differently from Candidates in prior papers. Rather than requiring both CEs to improve relative to a single baseline, it requires both CEs to lie between the two pure-component baselines. This is the correct prediction for within-class mixing: the mixed system should perform better than the worst pure component (DME, low SCE) and approach or match the better pure component (2-MeTHF, high SCE), while achieving the fixed point geometry. A result equal to or better than 2-MeTHF alone would indicate an additional contribution beyond the geometry-driven mechanism.

5 Candidate 2: LiFSI 1.0 M in FEME/2-MeTHF

5.1 FEME as a Fluorinated Linear Ether

Fluoroethyl methyl ether (FEME, $CH_2F-CH_2-O-CH_3$) is a fluorinated analogue of the linear ether class. The single fluorine substituent on the terminal carbon withdraws electron density from the ether oxygen, reducing the ESP minimum magnitude relative to DME and placing FEME between DME and highly fluorinated ethers in the coordination strength landscape.

The key property of FEME for this application:

$$|ESP_{\min}(\text{FEME}) - ESP_{\min}(\text{DME})| \approx 0.05\text{--}0.08 \text{ eV} \quad (7)$$

FEME coordinates Li^+ more weakly than DME. At equivalent concentrations, LiFSI/FEME electrolytes have higher CIP/AGG fractions than LiFSI/DME because the weaker solvent coordination allows more anion participation. SCE in LiFSI/FEME is therefore concentration- sensitive: increasing LiFSI concentration raises CIP/AGG fraction and raises SCE, while decreasing concentration lowers SCE toward theSSIP-dominant limit.

5.2 Concentration Tuning Equation

From Derivation 4 of the pre-registered derivation record, the SCE of LiFSI/FEME as a function of salt concentration c (mol L^{-1}) is fitted by:

$$\text{SCE}(\text{FEME}, c) = \text{SCE}_0 + \alpha \cdot c \quad (8)$$

where SCE_0 is the SSIP-dominant limit (extrapolated to zero concentration) and α is the concentration sensitivity coefficient. From the derivation record:

$$\text{SCE}_0 \approx 1.18, \quad \alpha \approx 0.19 \text{ L mol}^{-1} \quad (9)$$

At LiFSI 1.0 M:

$$\text{SCE}(\text{FEME}, 1.0) \approx 1.18 + 0.19 \times 1.0 = 1.37 \quad (10)$$

LiFSI/FEME at 1.0 M has $\text{SCE} = 1.37 < \text{SCE}^*$. The shell is too ordered for FEME alone to reach the fixed point by concentration tuning alone.

5.3 Can FEME Reach SCE^* by Concentration Alone?

Setting $\text{SCE}(\text{FEME}, c) = 1.466$:

$$c^* = \frac{1.466 - 1.18}{0.19} = \frac{0.286}{0.19} = 1.51 \text{ mol L}^{-1} \quad (11)$$

LiFSI 1.51 M in pure FEME would theoretically reach SCE^* by concentration alone. However, at this concentration:

- Viscosity increases substantially, reducing ionic conductivity.
- The CIP/AGG-driven SCE increase may overshoot SCE^* at slightly higher concentrations due to the steeply rising AGG fraction.
- The regime boundary may be crossed (CE_{RT} entering Regime 2 territory) depending on the AGG contribution.

A co-solvent approach using 2-MeTHF is therefore preferred: 2-MeTHF adds geometric diversity (cyclic vs. linear coordination modes) at a fixed salt concentration of 1.0 M, avoiding the viscosity and regime-crossing risks of high-concentration FEME alone.

5.4 Co-Solvent Requirement Calculation

From Derivation 4, the SCE of LiFSI 1.0 M in FEME/2-MeTHF as a function of 2-MeTHF volume fraction ϕ is estimated by:

$$\text{SCE}(\phi) \approx \text{SCE}(\text{FEME}, 1.0) + [\text{SCE}(\text{2-MeTHF}) - \text{SCE}(\text{FEME}, 1.0)] \cdot \phi = 1.37 + 0.18\phi \quad (12)$$

Setting $\text{SCE}(\phi) = 1.466$:

$$\phi^* = \frac{1.466 - 1.37}{0.18} = \frac{0.096}{0.18} = 0.53 \quad (13)$$

A 2-MeTHF volume fraction of $\phi^* \approx 0.53$ is estimated to reach SCE^* in LiFSI 1.0 M FEME/2-MeTHF. This corresponds to a volume ratio of approximately $\text{FEME:2-MeTHF} = 0.47 : 0.53 \approx 0.9 : 1$.

Remark 5.1. The FEME/2-MeTHF target ratio (approximately 0.9:1 FEME:2-MeTHF) is nearly equal-volume, in contrast to the 2-MeTHF/DME target ratio (1.7:1). The difference arises because FEME starts closer to SCE^* than DME does (SCE 1.37 vs. 1.24), requiring a smaller 2-MeTHF addition to reach the target. This difference is a prediction of the framework and is directly testable by the ratio scan protocol.

5.5 MU Session Confirmation

The MU session Query 3 (Master Document 8, same repository) confirmed:

- The solvation structure of LiFSI 1.0 M FEME/2-MeTHF at approximately 1:1 v/v: dominant FEME-linear coordination ($p_1 \approx 0.38$), secondary 2-MeTHF cyclic coordination ($p_2 \approx 0.37$), tertiary CIP minority ($p_3 \approx 0.25$).
- Computed SCE: $\text{SCE}_{\text{MU}} \approx 1.47 \pm 0.05$.
- Assessment: consistent with $\text{SCE}^* = 1.466$ within the uncertainty of the population estimate.
- The near-equal primary and secondary fractions are consistent with the within-class mechanism: FEME and 2-MeTHF are competing for the Li^+ shell with modest electrostatic preference for neither at the target ratio.

5.6 Formulation

Candidate 2: LiFSI 1.0 M in FEME/2-MeTHF, ratio to be confirmed by SCE measurement (estimated $\approx 0.9 : 1$ FEME:2-MeTHF).

Target SCE: 1.466 ± 0.05 .

Mechanistic basis: Within-class geometric diversity with fluorination-modified linear ether. FEME provides a fluorinated linear ether coordination geometry (weaker donor than DME, similar linear chain conformation). 2-MeTHF provides the cyclic monodentate geometry. At the target ratio, the population distributes across both geometries. The fluorination of FEME provides an additional tuning handle: FEME concentration can adjust SCE independently of the volume ratio via the concentration tuning equation.

5.7 Configuration Population Estimate

Table 2: Estimated Li^+ first-shell coordination geometry population for Candidate 2 at the target ratio (LiFSI 1.0 M, FEME/2-MeTHF $\approx 0.9:1$ v/v). Derived in Derivation 4 of the pre-registered record and confirmed by MU session Query 3.

Configuration	Symbol	Estimated fraction	Description
Dominant	p_1	≈ 0.38	FEME linear coordination (fluorinated chain, monodentate-like due to reduced bite)
Secondary	p_2	≈ 0.37	2-MeTHF cyclic monodentate (ring-constrained O- Li^+)
Tertiary	p_3	≈ 0.25	CIP minority (residual FSI $^-$ participation, FEME’s weaker coordination)
Computed SCE		$-\sum p_i \ln p_i$	$= 1.468 \approx 1.466 \checkmark$

5.8 Performance Predictions

$$\text{CE}_{\text{RT}}^{\text{pred}} = 100.1 - e^{1.493+1.230 \times 1.466} = 73.1\% \quad (14)$$

$$\text{CE}_{\text{LT}}^{\text{pred}} = 11.91 + 33.21 \times 1.466 = 60.6\% \quad (15)$$

These are Regime 1 geometry-driven predictions. FEME’s weaker Li^+ coordination may allow more film-forming SEI contributions from FSI^- decomposition than DME, potentially raising the experimental CE above the geometry-driven prediction.

5.9 Experimental Protocol

1. Prepare LiFSI 1.0 M solutions in FEME/2-MeTHF at volume ratios: pure FEME (baseline), 3:1, 1.5:1, 1:1, 0.7:1, 0.5:1 (FEME:2-MeTHF v/v).
2. Prepare a parallel concentration scan in pure FEME: LiFSI 1.0, 1.2, 1.4, 1.5 M. This tests the concentration tuning equation directly.
3. Measure Raman spectrum for each solution: FSI^- symmetric stretch ($\approx 740 \text{ cm}^{-1}$), $\text{Li}^+\text{-O}$ stretch ($300\text{--}500 \text{ cm}^{-1}$).
4. Compute SCE from population fractions. Identify the ratio at which $\text{SCE} \approx 1.466$. Compare to predicted $\approx 0.9 : 1$.
5. Also identify the concentration at which LiFSI/FEME pure reaches $\text{SCE} \approx 1.466$ and compare to the predicted $c^* = 1.51 \text{ M}$.
6. Measure CE_{RT} and CE_{LT} at the confirmed target ratio: 100 cycles, Cu/Li half cell, 0.5 mA cm^{-2} , 1.0 mAh cm^{-2} .
7. Apply falsification conditions.

5.10 Falsification Conditions

Pre-registered at <https://github.com/Eric-Robert-Lawson/attractor-oncology> before any experimental measurement.

- **Primary:** Framework falsified if CE_{RT} and CE_{LT} do not both improve relative to pure FEME at 1.0 M as 2-MeTHF fraction increases toward the target ratio.
- **Secondary:** Framework falsified if at the ratio confirmed by Raman to have $\text{SCE} \approx 1.466$: $\text{CE}_{\text{RT}} < 83\%$ OR $\text{CE}_{\text{LT}} < 59\%$.
- **Ratio prediction:** Falsified if the confirmed target ratio differs from $\approx 0.9 : 1$ by more than ± 0.4 (i.e., outside the range 0.5:1 to 1.3:1).
- **Concentration tuning:** Falsified if LiFSI/FEME pure does not reach $\text{SCE} \approx 1.466$ at a concentration within $\pm 0.2 \text{ M}$ of the predicted $c^* = 1.51 \text{ M}$.
- **Mechanistic:** Within-class mechanism falsified if Raman shows no shift in $\text{Li}^+\text{-O}$ modes between pure FEME and FEME/2-MeTHF at the target ratio.

Remark 5.2. Candidate 2 has five falsification conditions because it contains two independent predictions: the volume ratio prediction ($\phi^* \approx 0.53$) and the concentration tuning prediction ($c^* = 1.51$ M in pure FEME). These are derived from the same underlying equations and must both be confirmed for the full mechanistic picture to hold. Each constitutes an independent falsification test.

6 The Within-Class Mechanism Relative to the Cross-Class and Anion Receptor Mechanisms

Table 3: Comparison of the three SCE navigation mechanisms established across the series. The within-class mechanism occupies the middle ground in effect magnitude and practical accessibility.

Property	Cross-class ESP matching (Preprint 2, DME/TMP)	Anion receptor (Preprint 3, DOL/TMB)	Within-class geometric diversity (This paper, 2-MeTHF/DME, FEME/2-MeTHF)
Chemical classes involved	Ether + phosphate ester	Ether + Lewis acid (not coordination species)	Ether only
Source of SCE change	New coordination class (cross-class geometry)	Removal of anion participation from shell	New geometric sub-configuration within ether class
ESP contrast	$ \Delta\text{ESP} = 0.007$ eV ($< k_B T$)	Not applicable (anion receptor does not coordinate Li^+)	$ \Delta\text{ESP} \approx 0.03$ – 0.05 eV (1 – $2k_B T$)
SCE increase per unit fraction	Strong (cross-class contrast maximal)	Fine-tuning (small ΔSCE from near-approach)	Moderate (within-class contrast, partial preference)
Practical constraints	TMP flammability, phosphate ester stability	TMB volatility (bp 68°C)	None beyond standard ether handling
Established knowledge base	Partial (TMP is less common in Li metal)	Partial (TMB as additive, not primary mechanism)	Full (all major ether electrolyte literature applies)
Indicated when	Base solvent far below SCE^* in Regime 1, geometry-driven deficit	Base solvent has CIP/AGG-driven excess diversity	Practical ether-only formulation required; base solvent below SCE^*

The three mechanisms together define a complete practical toolkit:

- Maximum effect, cross-class: use ESP matching (Preprint 2).
- Anion-driven excess diversity: use anion reception (Preprint 3).
- Practical ether-only formulation: use within-class geometric diversity (this paper).

The selection criterion is the diagnostic: measure SCE, identify the primary source of its current value, select the mechanism that addresses that source. The repository index provides the complete navigation map.

7 DPE as a Negative Control: Concentration Invariance as a Predicted Non-Effect

Derivation 6 of the pre-registered derivation record demonstrates that dipropyl ether (DPE) is concentration-invariant with respect to SCE: increasing LiFSI concentration in DPE does not substantially change SCE because DPE’s coordination geometry is geometrically rigid and structurally uniform, and its long alkyl chains sterically exclude CIP/AGG formation at moderate concentrations.

$$\left. \frac{d(\text{SCE})}{dc} \right|_{\text{DPE}} \approx 0.02 \text{ L mol}^{-1} \quad (\text{vs. } \alpha_{\text{FEME}} \approx 0.19 \text{ L mol}^{-1}) \quad (16)$$

DPE therefore serves as a negative control for the concentration tuning mechanism: if the FEME concentration scan shows the predicted $\Delta\text{SCE} \approx 0.19$ units per mol L^{-1} increase, and a parallel DPE concentration scan shows $\Delta\text{SCE} \approx 0.02$ units per mol L^{-1} , the fluorination-driven concentration sensitivity of FEME is confirmed by contrast.

This comparison requires no additional experimental apparatus: it is a direct comparison of two solvents in the same salt/concentration scan protocol.

8 Discussion

8.1 The Significance of the Ratio Predictions

Both candidates in this paper carry specific numerical ratio predictions: 1.7:1 for Candidate 1 and $\approx 0.9 : 1$ for Candidate 2. These are not post-hoc fits. They are pre-registered predictions derived from the confirmed equations before any experiment. The ratio prediction falsification conditions test the framework’s quantitative accuracy, not just its directional accuracy.

A framework that correctly predicts only the direction (both CEs improve) but incorrectly predicts the ratio where SCE^* is achieved would indicate that the linear interpolation model requires a non-linear correction. That outcome would not falsify the framework’s core claim — the fixed point exists and is the target — but it would refine the interpolation model. The ratio predictions are therefore informative in all outcomes: confirmation, partial confirmation, and falsification each carry distinct mechanistic information.

8.2 The Within-Class Mechanism as the Practical Route

The cross-class ESP matching of Preprint 2 achieves the largest SCE shift per unit co-solvent fraction but introduces a phosphate ester into the formulation. The within-class mechanism trades that effect magnitude for complete ether-class compatibility.

The practical implication is direct: Candidates 1 and 2 are the candidates most immediately accessible to a laboratory without specialised solvent handling infrastructure. Both 2-MeTHF and DME are standard electrolyte solvents with established drying protocols, compatibility data, and CE benchmarks in the literature. FEME is commercially available and requires no unusual handling beyond standard anhydrous ether procedures.

This accessibility is not incidental. It is a design property of the within-class mechanism. A framework that can only be tested with exotic formulations is less useful than one that can be tested with standard reagents. Candidates 1 and 2 are the standard-reagent test of the framework.

8.3 The FEME Concentration Scan as an Independent Framework Test

The concentration tuning equation for FEME (Equation 8) is an independent quantitative prediction of the framework that is testable without any CE measurement. A concentration scan of LiFSI/FEME at 1.0, 1.2, 1.4, 1.51 M with Raman SCE estimation at each point tests whether the framework correctly predicts the concentration dependence of the solvation structure. This is a structural prediction, not a performance prediction. It is falsifiable without a single electrochemical measurement.

The predicted slope $\alpha = 0.19 \text{ L mol}^{-1}$ and the predicted intercept $\text{SCE}_0 = 1.18$ are both pre-registered and both testable by Raman. If confirmed, they constitute independent validation of the framework’s ability to quantitatively predict solvation structure from first principles.

8.4 Limitations

- The linear interpolation model for SCE as a function of volume fraction (Equation 2) is a first-order approximation. Non-linear effects (preferential solvation, activity coefficients, concentration-dependent coordination) will produce deviations from linearity, particularly at extreme ratios. The ratio scan protocol is designed to map the actual SCE–ratio curve and identify deviations.
- The FEME concentration tuning parameters ($\text{SCE}_0 = 1.18$, $\alpha = 0.19 \text{ L mol}^{-1}$) are derived from a limited dataset. They should be treated as estimates with $\pm 20\%$ uncertainty until confirmed by experiment.
- 2-MeTHF is known to undergo ring-opening at Li metal surfaces, forming a flexible SEI. This film-forming chemistry is separate from the SCE mechanism and may produce a CE contribution that is not predicted by the framework. Experiments should include a film-forming control (e.g., comparing electrodes pre-treated with 2-MeTHF SEI vs. fresh electrodes).
- FEME is a less commonly characterised solvent in Li metal electrolytes than DME or 2-MeTHF. Its reductive stability at Li metal potentials should be confirmed before extended cycling.
- The regime classification of both candidates must be confirmed experimentally. If CE_{RT} exceeds 90% at the target ratio, the Regime 2 equations apply and the interpretation changes.

9 Conclusion

We have introduced **within-class geometric diversity** as the third rationally designed mechanism for navigating toward $\text{SCE}^* = 1.466$ in Li metal battery electrolytes.

The mechanism operates as follows: two solvents from the same chemical family (the ether class) but with structurally distinct Li^+ coordination geometries (cyclic monodentate vs. linear bidentate, or fluorinated linear vs. cyclic) are mixed such that both coordination modes are simultaneously and meaningfully populated in the Li^+ primary shell. SCE rises toward SCE^* . The effect is weaker per unit volume fraction than the cross-class ESP-matching mechanism because the geometric contrast is smaller, but the practical accessibility is higher because no cross-class co-solvents are required.

Two candidates are derived, pre-registered, and fully specified:

- **Candidate 1:** LiFSI 1.2 M in 2-MeTHF/DME, 1.7:1 v/v. Predicted SCE = 1.466 confirmed by MU session Query 1 ($\text{SCE}_{\text{MU}} \approx 1.46 \pm 0.04$). Four falsification conditions pre-registered.

- **Candidate 2:** LiFSI 1.0 M in FEME/2-MeTHF, $\approx 0.9 : 1$ v/v. Predicted SCE = 1.466 confirmed by MU session Query 3 ($\text{SCE}_{\text{MU}} \approx 1.47 \pm 0.05$). Five falsification conditions pre-registered, including an independent concentration tuning test.

DPE is identified as a negative control for the concentration sensitivity mechanism: its predicted concentration invariance ($\alpha \approx 0.02 \text{ L mol}^{-1}$) contrasts with FEME’s concentration sensitivity ($\alpha \approx 0.19 \text{ L mol}^{-1}$) and is directly testable.

The three mechanisms now established across the series — cross-class ESP matching, anion reception, and within-class geometric diversity — constitute a complete practical toolkit for rational navigation to SCE* from any starting formulation in the ether class. The full geometric structure connecting all mechanisms, candidates, derivations, and the fixed point is navigable from the repository index at <https://github.com/Eric-Robert-Lawson/attractor-oncology>.

Data and derivation record availability. All pre-registered predictions, falsification conditions, population estimates, interpolation derivations, concentration tuning equations, and MU session records referenced in this paper are archived with full timestamped commit history at:

<https://github.com/Eric-Robert-Lawson/attractor-oncology>

The repository commit history constitutes the priority and pre-registration record for all claims in this paper.

Relationship to other papers in this series. This is Preprint 4 of the SCE Framework series. Preprints 1, 2, and 3 are available at the same repository. All papers in the series are self-contained. The geometric structure connecting all papers, mechanisms, candidates, and derivations is navigable from the repository index.

Competing interests. The author declares no competing financial interests.

Author contributions. E.R.L.: conceptualisation, derivation, interpolation modelling, concentration tuning analysis, writing.

A Solvent Physical Properties

Table 4: Physical properties of solvents used in Candidates 1 and 2.

Solvent	mp ($^{\circ}\text{C}$)	bp ($^{\circ}\text{C}$)	ρ (g mL^{-1})	DN	SCE (LiFSI 1M)
2-MeTHF	−137	80	0.854	20	1.55
DME	−58	85	0.867	24	1.24
FEME	−90	72	0.902	≈ 18	≈ 1.37
DPE (control)	−122	90	0.736	19	1.28

FEME properties from [10, 2]. DN values from [3]. SCE values from Preprint 1 dataset (2-MeTHF, DME, DPE) and Derivation 4 estimate (FEME).

B Derivation of the Corrected Target Ratio for Candidate 1

The linear interpolation estimate gives $\phi^* = 0.729$ (2-MeTHF volume fraction), corresponding to 2-MeTHF:DME = 2.69:1 at LiFSI 1.0 M. At LiFSI 1.2 M, increased salt concentration raises the SCE of the pure 2-MeTHF component modestly (CIP/AGG fraction increases in concentrated solutions) while raising the pure DME component SCE by a similar amount. The net effect on the

interpolated target ratio is a shift toward lower 2-MeTHF fraction, because the added CIP/AGG fraction in DME at 1.2 M salt partially substitutes for the cyclic-geometry diversity that 2-MeTHF would otherwise provide.

Estimated correction: $\Delta\phi \approx -0.10$, giving $\phi_{\text{corrected}}^* \approx 0.629$, corresponding to 2-MeTHF:DME $\approx 1.7 : 1$.

Full algebraic derivation is in the pre-registered record at <https://github.com/Eric-Robert-Lawson/attractor-oncology>, file `derivation_for_novel_result.md`, Derivation 3.

C Pre-Registered Falsification Conditions (Verbatim)

Candidate 1 (LiFSI 1.2 M, 2-MeTHF/DME 1.7:1 v/v):

- *Primary*: Both CEs lie between DME and 2-MeTHF pure-component baselines and both exceed DME baseline simultaneously.
- *Secondary*: At Raman-confirmed SCE ≈ 1.466 : $\text{CE}_{\text{RT}} \geq 84\%$ AND $\text{CE}_{\text{LT}} \geq 60\%$.
- *Ratio*: Raman-confirmed target ratio within ± 0.5 of 1.7:1.
- *Mechanistic*: Li^+ -O Raman shift observed between pure 2-MeTHF and 1.7:1 mixed system.

Candidate 2 (LiFSI 1.0 M, FEME/2-MeTHF $\approx 0.9:1$ v/v):

- *Primary*: Both CEs improve relative to pure FEME 1.0 M as 2-MeTHF fraction increases toward target ratio.
- *Secondary*: At Raman-confirmed SCE ≈ 1.466 : $\text{CE}_{\text{RT}} \geq 83\%$ AND $\text{CE}_{\text{LT}} \geq 59\%$.
- *Ratio*: Confirmed target ratio within ± 0.4 of 0.9:1.
- *Concentration tuning*: Pure FEME reaches SCE ≈ 1.466 within ± 0.2 M of predicted 1.51 M.
- *Mechanistic*: Li^+ -O Raman shift observed between pure FEME and FEME/2-MeTHF at target ratio.

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